Namal University Transport Management System

Database Project Proposal

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0.1 Introduction

Namal University, Mianwali, provides transport services to students, faculty, and staff for daily commutes between the campus, hostels, and nearby city areas. Currently, these operations are managed through manual, paper based processes that lead to errors, delays, and lack of transparency. Administrators and drivers benefit from a centralized platform to manage routes, monitor trips, and generate accurate reports, improving overall efficiency and user convenience.

The Namal University Transport Management System (NUTMS) is a web based platform that aims to digitize and automate these transport operations. This Database Systems proposal focuses on the database design and management layer of NUTMS the backbone that will store, manage, and retrieve all transport related data reliably and efficiently. A well designed relational database is critical to NUTMS because it supports every core operation : user registration, GPS based real time tracking, route and schedule management, driver assignments, notification delivery, and analytics reporting.

This proposal outlines the problem being addressed, the system objectives, scope, data requirements, technology stack, functional capabilities, user roles, and the expected impact of the database system on university transport operations. The database will be developed using MySQL, which is a widely used and reliable relational database management system that suits the structured nature of transport data. It will ensure that all records are stored consistently, retrieved quickly, and remain protected from unauthorized access. By building a strong database foundation, NUTMS aims to make university transport more organized, transparent, and easy to manage for everyone involved.

0.2 Problem Statement

Namal University, Mianwali, currently manages its transport operations through manual, paper based processes. There is no centralized database system to store, manage, and retrieve transport related data in a reliable and organized way. All information related to routes, schedules, drivers, and vehicles is maintained through physical registers and informal spreadsheets, with no single place where all transport data can be stored, accessed, and managed efficiently. Since records are maintained manually by different people at different times, the same data often exists in conflicting forms, leading to data inconsistency across the system. Students and faculty have no way to check current bus locations, schedule changes, or delays through any system, as all updates are communicated through phone calls or any other informal way which is unreliable and slow. Furthermore, transport administrators spend excessive time manually preparing reports and managing driver and vehicle assignments, which increases the chance of human error and reduces overall efficiency. The proposed MySQL based database system for NUTMS directly addresses all of these gaps by providing a structured, consistent, and secure data management solution for all university transport operations.

0.3 Objectives

The NUTMS database system aims to achieve the following measurable and clearly defined goals :

1. **Store and Retrieve Real Time GPS Data :** Design a GPS logs table capable of storing location updates every 10 to 15 seconds per bus, with indexing on bus ID and timestamp so that the latest bus location can be retrieved within 5 seconds.

2. **Implement Role Based Access Control :** Define separate access levels in the database for Students, Faculty, Drivers, and Administrators so that each user can only access the data and functions that their role permits.

3. **Maintain Complete Audit Logs :** Record all critical user actions including bookings, cancellations, approvals, login attempts, and profile updates with user ID, action type, and timestamp to ensure full accountability and transparency.

4. **Support Report Generation :** Enable the database to efficiently run analytical queries that generate daily, weekly, and monthly reports on transport usage, peak hours, route performance, and driver trip completion rates.

5. **Ensure Data Availability and Security :** Maintain 99% database availability during academic semesters through cloud hosting, implement daily automated backups with 30 day retention, and encrypt all sensitive user data to prevent unauthorized access.

0.4 Scope

The Namal University Transport Management System (NUTMS) is a web based platform designed specifically to manage and automate the transport operations of Namal University, Mianwali. The scope of this system defines the boundaries of what the database will store, manage, and support. The following features are included within the scope of this project:

User Management: The system will store and manage the profiles of all users including students, faculty, drivers, and administrators. Each user will have a defined role with specific access levels and permissions.

Route and Schedule Management: The database will store all bus routes, stop locations, departure times, and arrival times. Administrators will be able to add, update, and manage routes and schedules through the system.

Real Time GPS Tracking: The system will store real time location data received from GPS devices installed in university buses. Students and faculty will be able to view the current location of buses on a map.

Driver and Vehicle Management: The database will maintain complete records of all drivers and vehicles including vehicle capacity, registration details, and driver assignments to specific routes.

Notification System: The system will store and manage notifications sent to users regarding bus delays, schedule changes, and emergency alerts through email.

Special Vehicle Booking for Faculty: Faculty members will be able to submit requests for special vehicle bookings for trips. Administrators will review and approve or reject these requests through the system.

Administrative Dashboard and Reporting: The system will provide administrators with a centralized dashboard to monitor transport operations and generate reports on route usage, driver performance, and transport efficiency.

Audit Logs: The database will maintain a complete record of all important user actions such as bookings, cancellations, approvals, and login attempts for accountability and transparency.

0.5 Data Collection

Data collection for Namal Transport Management Systems (NUTMS) will be comprehensive to ensure effective management. Different data sources include university records for students and faculty including their emails, Name, and reg ID, transport logs for historical route, buses schedule, and real time address for GPS tracking.

0.5.1 Students & Faculty Data

The data for the student & faculty will be obtained from university's central information systems (SSO & Admission Office) which includes their enrollment status, contact and other details, academic schedule and residential address for day scholars.

0.5.2 Transport

The data for transport includes buses routes, departure schedules, vehicle information i.e plate number, and driver assignments which will be gathered from transport department's manual record.

0.5.3 Drivers and Vehicle Management Data

Driver and vehicle management data will be collected from transport department files and records including drivers information, fuel consumption and license information.

0.5.4 Global Positioning System

GPS will access will be given access for real time location of the routes where performance will be captured through API integration with GPS tracking hardware and web application.

Each step will follow a structured plan with data validation to ensure accuracy, and consistency before uploading to any database. Initial data population will be performed through secure CSV imports and SQL scripts, followed by continuous real-time data ingestion through application APIs.

0.6 Technology Stack

The NUTMS will utilize a modern and scalable technology stack optimization for the application.

- **SQL:** This core database management system will used for its ACID compliance, advance indexing capabilities, and proven reliability in high-transaction environments.SQL Server's robust relational model ensures data integrity through primary/foreign key constraints, while its full-text search capabilities enable efficient route and schedule queries. The DBMS supports complex JOIN operations across multiple normalized tables for comprehensive transport analytics and reporting.

- **Backend:** Node.js and Express,js framework will be used to create RESTful APIs and WebSocket for real-time GPS updates and notification. React.js with TypeScript will power responsive frontend, for consistent user experience across desktop, tablet and mobile devices. Socket.io will handle real-time bidirectional communication for live bus tracking and instant notifications.

- **OpenStreetMap:** For geospatial visualization, Leaflet.js with OpenStreetMap tiles will provide cost-effective, high-performance mapping without vendor lock-in. Nodemailer with university SMTP integration will manage automated email notifications, while JWT (JSON Web Tokens) will secure user authentication and role-based access control.

- **Containerization:** The application will be containerized using Docker for production environment while Git hub will implement CI/CD pipelines for automated testing deployment to Vercel cloud platform. PgAdmin 4 will serve as the database administration tool, and Postman will facilitate API testing and documentation.

Following technology stack ensures scalabilty, security and cost-effectiveness which meets all functional requirements of the system.

0.7 System Functionality

NUMTS is delivering scalable functionalities across following areas to fully automate the university's transport operations:

1. **User Registration and Authentication:** The system will secure registrations for students, faculty, drivers, and administrators with role-based control. System provides multi-factor authentication for security and password reset functionality.

2. **Route management:** Route definition and maintenance system allow administrator to create, edit or optimize bus transportation routes. The system allow administrator to define routes by specifying sequential stops with exact locations. Each routes include metadata such as route name, vehicle capacity or notes for drivers.

3. **Schedule management:** NUTMS allow administrators to manage daily, and weekly transport planning based on university's semester classes schedule. This prevents scheduling conflicts such as buses departure after a specific slot. Students can also get notification for each bus departure.

4. **Real-time Tracking:** The system is integrated with GPS for continuous location updates which allow user to track estimated time calculation for destination. OpenStreetMap provide interactive map visualization which make it easy to track routes.

5. **Faculty Booking:** System allow faculty to request special trips which requires a submission form, administrative approval process and dedicated vehicle assignments.

6. **Notification System:** Automated notification are enabled for buses departure or arrival, updates in buses schedule or changes, and delays. In case of an emergency, emergency alerts can be sent to administrative department for rapid action.

7. **Administrative Control:** System allows administrators to update or edit manual operations, such as system audits log , override capabilities and schedule update. Administrators can view a comprehensive report on weekly and monthly bases.

8. **Reporting and Analytic:** System generates report on weekly and monthly bases which covers transport utilization, route efficiency, driver performances, weekly fuel consumption and data visualization through interactive charts and graphs.

These functionalities replaces a manual paper based system to an automated system ensuring user satisfaction, daily work efficiency, and resource utilization. NUMTS boosts security measures with data-driven decision making.

0.8 User Classes

The Namal University Transport Management System (NUTMS) is designed to support four main types of users. Each user class has a different role and a defined level of access to the system based on their responsibilities.

Students: Students are the primary users of the system. They can log in to view bus routes, schedules, and real time bus locations on the map. They can also receive notifications about bus delays or schedule changes. Students can only view transport information relevant to them and cannot modify any data in the system.

Faculty: Faculty members have the same access as students for viewing routes, schedules, and bus locations. In addition, faculty members can submit special vehicle booking requests for trips outside the regular transport schedule. They can view the status of their submitted requests and receive notifications about approval or rejection from the administrator.

Drivers: Drivers can log in to view their assigned routes, schedules, and passenger information for their trips. They can update the trip status such as started, delayed, or completed so that the system can display accurate real time information to other users. Drivers cannot access booking records, administrative functions, or other users data.

Transport Administrators: Administrators have full access to the system. They can manage (add, delete and update) all users, buses, drivers, routes, and schedules. They can approve or reject special vehicle booking requests submitted by faculty, send emergency notifications to users, monitor all transport operations through the admin dashboard, view audit logs, and generate reports on transport performance and usage. Administrators are responsible for the overall management and smooth operation of the transport system.

Each user class is assigned a specific role in the database with permissions that restrict access to only the data and functions that are relevant to their responsibilities. This ensures data security and prevents unauthorized access to sensitive information.

0.9 Task Division

Team Member	Responsibilities
Altaf Hussain	Wrote the Introduction, Task Division, Conclusion, and References sections of the proposal report. Managed the GitHub repository by uploading and maintaining all project files. Compiled and integrated all sections written by team members into the final report and ensured the document was properly formatted and submitted on time.
Rabia Ashraf	Wrote the Problem Statement, Objectives, Scope, and User Classes sections of the proposal report. Identified the key problems in the current manual transport system, defined measurable objectives for the database system, specified the system boundaries, and described the roles and access levels of all user classes.
Najia Nayab	Wrote the Data Collection, Technology Stack, and System Functionality sections of the proposal report. Identified data sources and collection methods, justified the selection of MySQL and supporting technologies, and outlined the key functional requirements that the database system will provide to its users.

0.10 Conclusion

This proposal presents the database system design foundation for the Namal University Transport Management System (NUTMS), building upon the Software Engineering project proposal and SRS document previously developed by the same group.

The proposed MySQL based relational database will replace Namal University's current manual, paper based transport management processes with a centralized, reliable, and scalable data management solution. By implementing role based access control, storing real time GPS data, and maintaining comprehensive audit trails, the system will directly address the critical operational challenges identified in the problem statement.

The expected impact of the NUTMS database system includes : real-time transparency for students and faculty through live GPS data queries; significant reduction in administrative overhead through automated reporting and centralized data management; improved safety compliance through enforced capacity limits; and data driven decision making enabled by analytics queries on historical transport data.

By using cost effective, open source technologies (MySQL, Node.js, React.js) the system will be sustainable within the university's resource constraints while meeting all functional and non-functional requirements specified in the SRS.

This proposal establishes the foundation for the design and implementation of the database system in subsequent project milestones.

0.11 References

1. NUTMS Software Engineering Project Proposal (2025). Prepared by Altaf Hussain, Rabia Ashraf, Najia Nayab. Namal University, Mianwali.

2. NUTMS Software Requirements Specification (2025). Prepared by Altaf Hussain, Rabia Ashraf, Najia Nayab. Namal University, Mianwali.

3. OpenAI. (2025). ChatGPT (GPT-4o) [Large language model]. Accessed March 2026.

4. Creatrix Campus. (n.d.). Transport Management System for colleges. Retrieved March 12, 2026, from https://www.creatrixcampus.com/transport-management-system

5. Claude. (2026). Claude (Claude-Sonnet-4). Accessed March 2026.

6. GitHub Repository
   `https://github.com/Altaf-Hussain-11/DB-Project-NUTMS-.git`

Appendix A: AI Tools Used

AI Tool	Purpose	Prompts / Usage Description	Access Date
ChatGPT	Concept understanding and LaTeX Code	Used prompts : "Can you explain what each section of a proposal report means and what kind of information is generally written in each one?" and "What are the common problems faced in manual university transport management and how can a database system solve them?"	Feb–Mar 2026
ChatGPT	To explore and compare different databases and choose the most appropriate and suitable one	Above I have uploaded the Proposal and SRS of our Software Engineering Project. We are continuing the same project in the database. We are developing Namal University Transport Management System that handles user registration, seat booking, real-time GPS tracking, route and schedule management, driver assignments, notification delivery, and analytics reporting. The system will have multiple concurrent users including students, faculty, drivers, and administrators. Which relational database management system would be most suitable for this project and why? Also tell why others are not suitable for this.	Feb–Mar 2026
Claude	Document formatting	We are working on a Database Systems course project. We need to write a project proposal for our project. The milestone requires the following sections to be covered in the report : Introduction, Problem Statement, Objectives, Scope, Data Collection, Technology Stack, System Functionality, User Classes, Task Division, Conclusion, References, and an AI Prompts Appendix. The proposal should be professionally structured, properly formatted, and written in simple and clear academic language suitable for our report. Provide a clean document template with proper headings, subheadings, and formatting structure for each section so we can fill in our own content.	March 2026
ChatGPT	To generate LaTeX code for document structure, formatting, tables, and layout	Convert this text into a properly formatted LaTeX document with title page, table of contents, section headings, and professional academic styling.	March 2026